

Drew Halverson

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EXPERIENCE

Software Engineer

March 2025 – Present

Google

Sunnyvale, CA

- Optimized core validation service database reads for burst heavy workloads, achieving a ~5.5x speedup; reduced worst case execution time for 100,000 entry streaming RPCs from 71 to 13 minutes while eliminating transaction aborts and lowering **Spanner DB** contention
- Took ownership of the core validation service, re-engineering it into an extensible framework to support complex validations across different **Spanner DB** tables
- Rearchitected tightly coupled **RPC** interactions across **distributed systems** by removing fragile flag based logic and consolidating network round trips, improving overall maintainability and operational visibility
- Implemented several edge networking products in the Network Product Ordering system, allowing business critical infrastructure to be ordered and tracked in an automated fashion

Software Engineer

July 2022 – February 2025

Palo Alto Networks

Santa Clara, CA (Remote)

- Designed and deployed a reverse proxy in **Go** on **Google Kubernetes Engine** to shard tenant specific API requests across distributed systems, removing architectural limits on tenant capacity
- Took end to end ownership of a core **Go** and **gRPC** scheduling service, refactoring **Temporal** workflows to achieve a ~5x increase in agent capacity per tenant
- Removed need for multiple joins during a frequent query to **MongoDB**, reducing query times by ~50%

Software Development Intern

June 2021 – August 2021

Amazon

Seattle, WA (Remote)

- Streamlined deployment of datasets used to inform delivery drivers on tasks such as locating a mailroom, unlocking a door with a code, or finding the correct unit of an apartment/condo complex
- Developed **Java** code on **AWS Lambdas** to monitor deployment progress tracked in **DynamoDB**, used **SQS** to communicate these events between Lambdas and **AWS AppConfig** to serve newest versions to the mobile app

Software Engineer Team Lead

October 2020 – April 2021

Trulight Project

Cedarburg, WI (Remote)

- Led a team of interns working on backend algorithm and API development; organized the team's work using **Jira**, prioritized critical tasks for release of first major version of the backend

Software Engineer Intern

March 2020 – October 2020

Trulight Project

Cedarburg, WI (Remote)

- Developed APIs hosted on **GCP App Engine**, **Cloud Run** and **Endpoints** with **Java Spring** framework that aggregates multiple external APIs' data to calculate a TruLight safety score for the user's location

PROJECTS

MugShot - Coffee Tracker | *Go, PostgreSQL, GCP, React*

December 2025 - Present

- Engineered a full stack social media platform enabling users to rate, share, and explore coffees and cafes
- Architected a **concurrency safe Go** backend using a modular layered design with custom middleware for auth, rate limiting, CORS, and pagination
- Containerized and deployed the **Go** API to **GCP Cloud Run** backed by **PostgreSQL** and **Google Cloud Storage**, with automated testing and deployment via **GitHub Actions**

Fetch Receipt Processor | *Go*

June 2024 - July 2024

- Engineered a concurrency safe **Go microservice** exposing **RESTful APIs** to process, store, and retrieve receipts

TECHNICAL SKILLS

Languages: Go, Java, Python, C/C++, SQL

Databases: Google Cloud Spanner, PostgreSQL, MongoDB, DynamoDB

Infrastructure & Systems: GCP, AWS, GKE, Docker, Temporal, gRPC, Protobuf, Grafana, Lightstep, CI/CD

EDUCATION

University of Wisconsin - Madison

May 2022

Bachelor of Science in Computer Science – GPA: 3.67